BUYER'S VS. SELLER'S MARKET CLASSIFIER

PROBLEM

- Real estate investment firms need a scalable and data-driven way to identify which of 900+ U.S. metros favor buyers vs. sellers to prioritize deal-sourcing as existing tools like Zillow's Market Heat Index report a single score per metro without an actionable signal, and manually tracking dozens of indicators across hundreds of markets doesn't scale.

GOALS AND OBJECTIVES

- **Classify each of 900+ U.S. metros monthly into a Buyer's/Balanced/Seller's Market label** via an end-to-end ML pipeline that corrects for the housing market's broader boom-bust cycle, so labels reflect a metro's relative character rather than which year it is.

ACTION

- **Integrated five Zillow Research datasets** (Market Heat Index, Mean Days to Pending, Share of Listings with a Price Cut, Sale-to-List Ratio, ZHVI) by unpivoting each from wide to long format and loading them into SQLite, joining on RegionID and Date across ~900 U.S. metros and 8 years of monthly data via multi-table SQL joins.
- **Diagnosed structural data-quality issues**, tracing non-random missingness in two candidate features to Zillow's minimum sales-volume reporting threshold, and identifying a data-leakage risk in a third feature by cross-referencing Zillow's own published methodology.
- **Engineered a leakage-free, relative target and feature set** — a 3-class target using within-month percentile binning (correcting for the 2020–2022 boom-bust cycle), plus percentile-rank versions of key predictors (correcting a feature-target mismatch that had caused systematic bias).
- **Trained a multinomial logistic regression classifier** on a time-based train/test split, validated feature independence via VIF, benchmarked against K-means, and interpreted coefficients as odds ratios.

RESULT

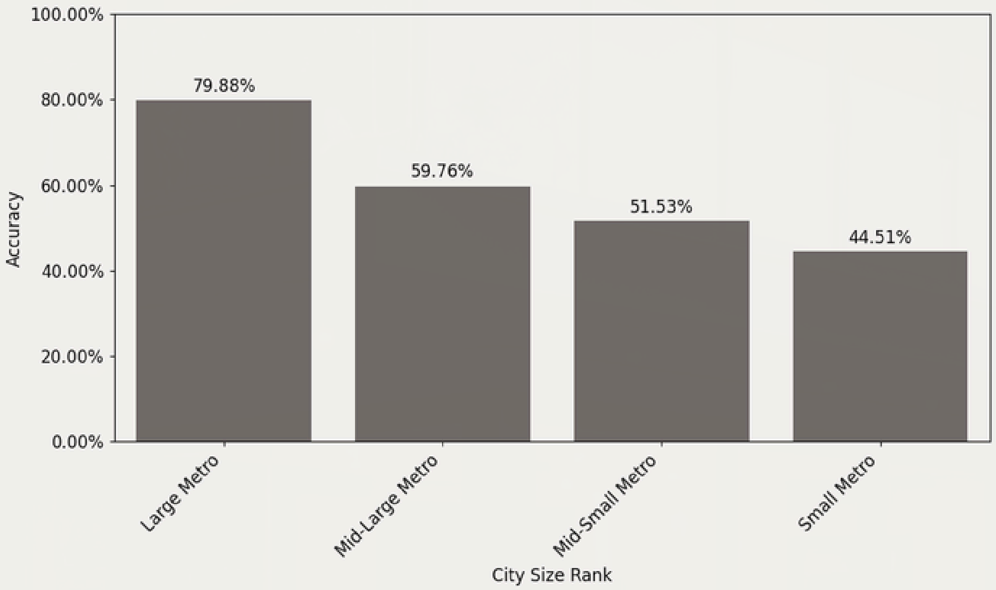
- **Achieved 62% accuracy** on held-out test data (versus a 33% random-guess baseline for three classes), with feature engineering measurably correcting an early systematic bias toward over-predicting Buyer's Markets.
- **Validated generalization on a genuinely out-of-sample month**; performance stayed strong for large, liquid metros but degraded for smaller markets — a limitation traced to Zillow's own reporting thresholds, not a modeling flaw.
- **Identified Sale-to-List Ratio as the strongest predictor**, where a one-standard-deviation increase in a metro's relative sale-to-list ratio roughly tripled the odds of a Seller's Market classification.

Exhibit 1: Feature Engineering Impact — Model v1 vs. v2 (table)

Metric	Model v1 (raw features)	Model v2 (percentile-rank features)
Overall Accuracy	60%	**62%**
Seller's Market Recall	47%	**75%**
Buyer's Market Precision	56%	**64%**
Macro Avg F1-Score	0.58	**0.61**

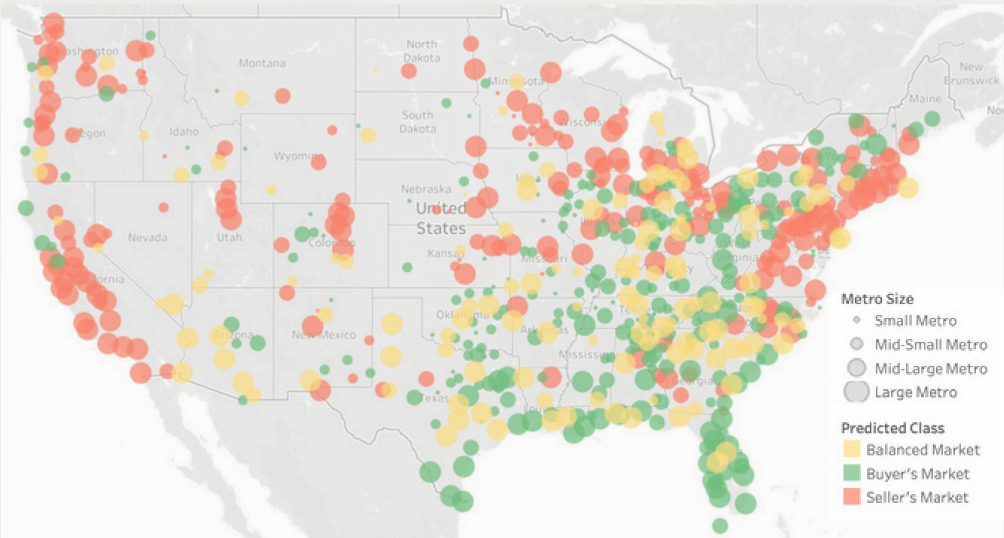
Percentile-rank feature engineering lifted overall accuracy from 60% to 62%, with the largest gain in Seller's Market recall (47% → 75%), correcting the earlier model's tendency to miss true seller's markets.

Exhibit 2: Held-Out Prediction Accuracy by Metro Size (chart)



Accuracy on the most recent, held-out month ranges from 79.88% for the largest metros to 44.51% for the smallest (a gap traced to Zillow's own data-reporting thresholds for low-volume markets, not a modeling flaw).

Exhibit 3: Predicted U.S. Housing Market Conditions (April 2026)



Model-predicted classification across 900+ U.S. metros for April 2026, the most recent month in the dataset. Circle size reflects relative metro population; color indicates predicted market type.